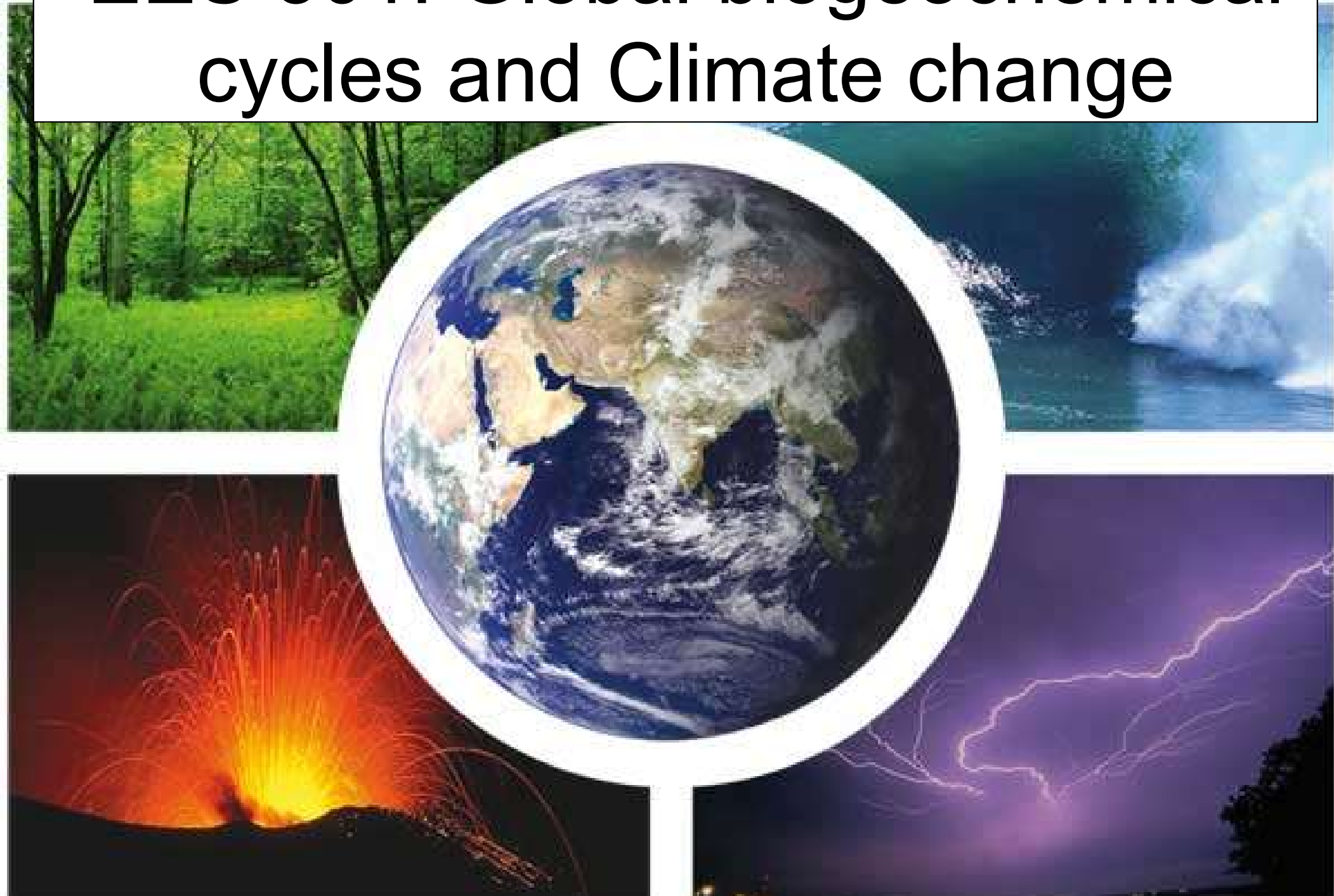
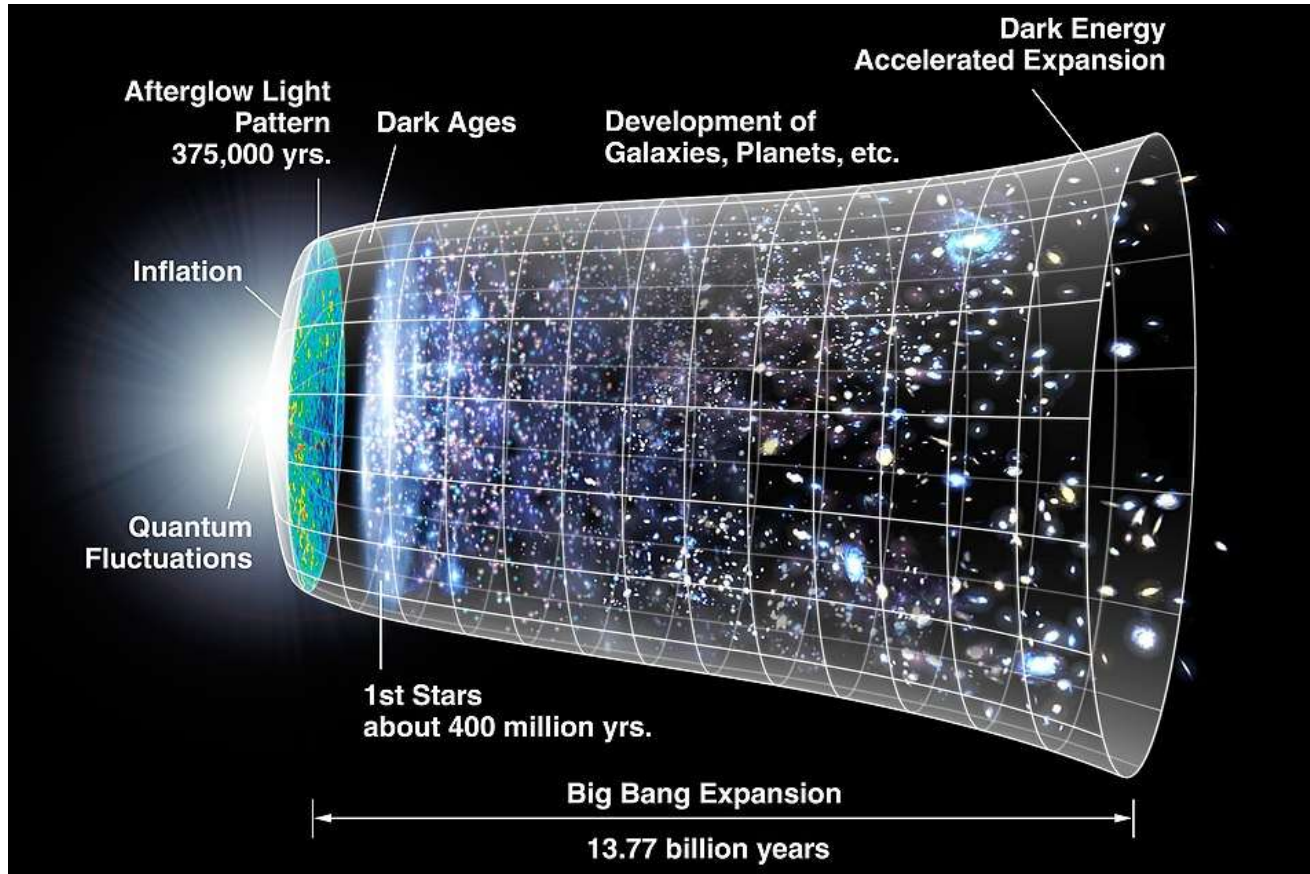


EES 301: Global biogeochemical cycles and Climate change



“It all started with a big bang...”

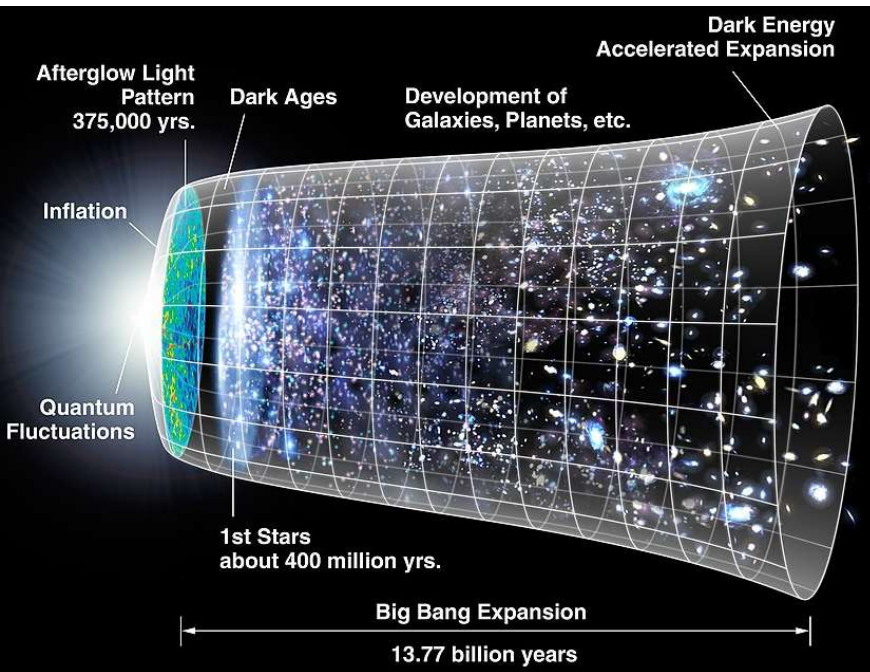


~13.8 billion years ago, a massive blast allowed all the universe's known matter and energy—even space and time themselves—to spring from some ancient and unknown type of energy.

The theory maintains that, in the instant—a trillion-trillionth of a second—after the big bang, the universe expanded with incomprehensible speed from its pebble-size origin to astronomical scope. Expansion has apparently continued, but much more slowly, over the ensuing billions of years.

https://en.wikipedia.org/wiki/Big_Bang

Stages of the Big Bang



1. ~Superfast inflation from size of atom to size of a grapefruit in a fraction of a second – the big bang marks the beginning of space-time
2. Post inflation the universe is a seething hot soup of electrons, quarks and other particles – nowadays high energy physics studies these with particle colliders
3. A rapid cooling permits quarks to clump into protons and neutrons
4. At 3 minutes after the Big Bang the Universe is still too hot to for atoms to form. Charged electrons and protons prevent light from shining. The universe s a superhot fog. The remnants of the fog are seen l the cosmic microwave background radiation
5. At 300,000 years after the Big Bang the Universe has cooled enough for electrons to combine with protons and neutrons to form atoms. Mostly hydrogen and helium, light can finally shine.

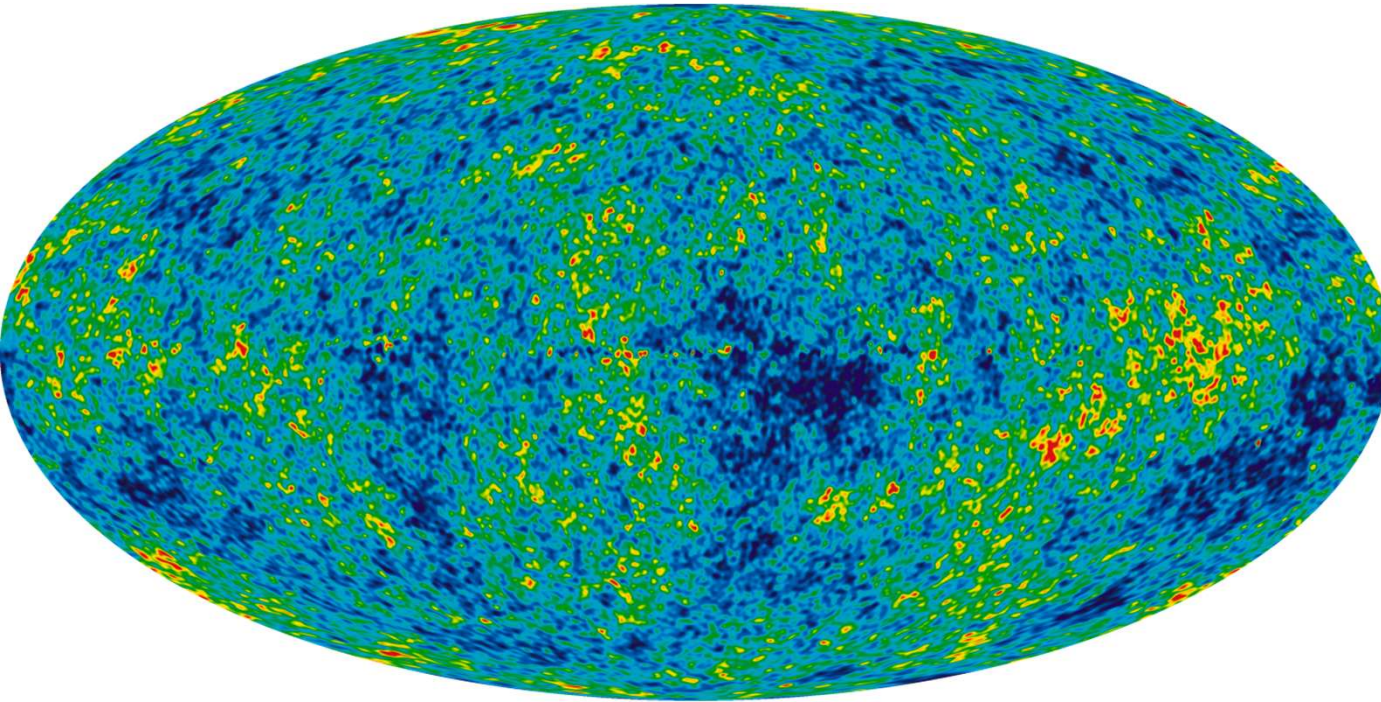
https://en.wikipedia.org/wiki/Big_Bang

6. Gravity makes hydrogen and helium gas coalesce to form the giant clouds that will become galaxies. Smaller clumps of gas collapse onto themselves to form the first stars.
7. As galaxies cluster together under gravity the first stars die and spew heavy elements into space. Those will eventually turn into new stars and planets. Stars can only make elements up to iron by fusion. All heavier elements are made by neutron capture primarily in the blast that throws out elements, radioactive radiation and neutrons during supernova explosions.

The development of the Big Bang Theory

- A Belgian priest named **Georges Lemaître** first suggested the big bang theory in the **1920s** when he theorized that the universe began from a single primordial atom. He derived the expansion of the Universe from the equations of general relativity in an **1927** article and proposed a first value for the Hubbles constant.
- The idea subsequently received major boosts by Edwin Hubble's observations that galaxies are speeding away from us in all directions. **In 1929, from analysis of galactic redshifts, Edwin Hubble concluded that galaxies are drifting apart;** this is important observational evidence consistent with the hypothesis of an expanding universe. Hubble found that the magnitude of the redshift is largest of galaxies furthest away from the Milky Way (our galaxy)
- **The cosmic microwave background (CMB) radiation was discovered accidentally in 1964 by Arno Penzias and Robert Wilson.** They were working with a sensitive radio telescope at Bell Labs when **they detected a persistent, low-level noise coming from all directions in the sky.** Initially, they attributed it to interference, but further investigation revealed it to be a faint afterglow of the Big Bang, now known as the CMB

The Goldilocks condition for the formation of Galaxies and stars were...



Tiny fluctuations in the temperature and density of the material in the early universe that can be seen in the cosmic background radiation

By NASA / WMAP Science Team -

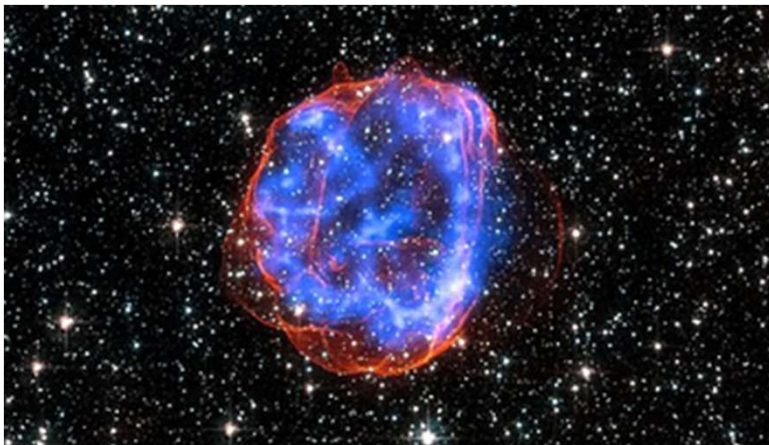
http://map.gsfc.nasa.gov/media/121238/ilc_9yr_moll4096.png, Public Domain,

<https://commons.wikimedia.org/w/index.php?curid=23285693>

The creation of elements



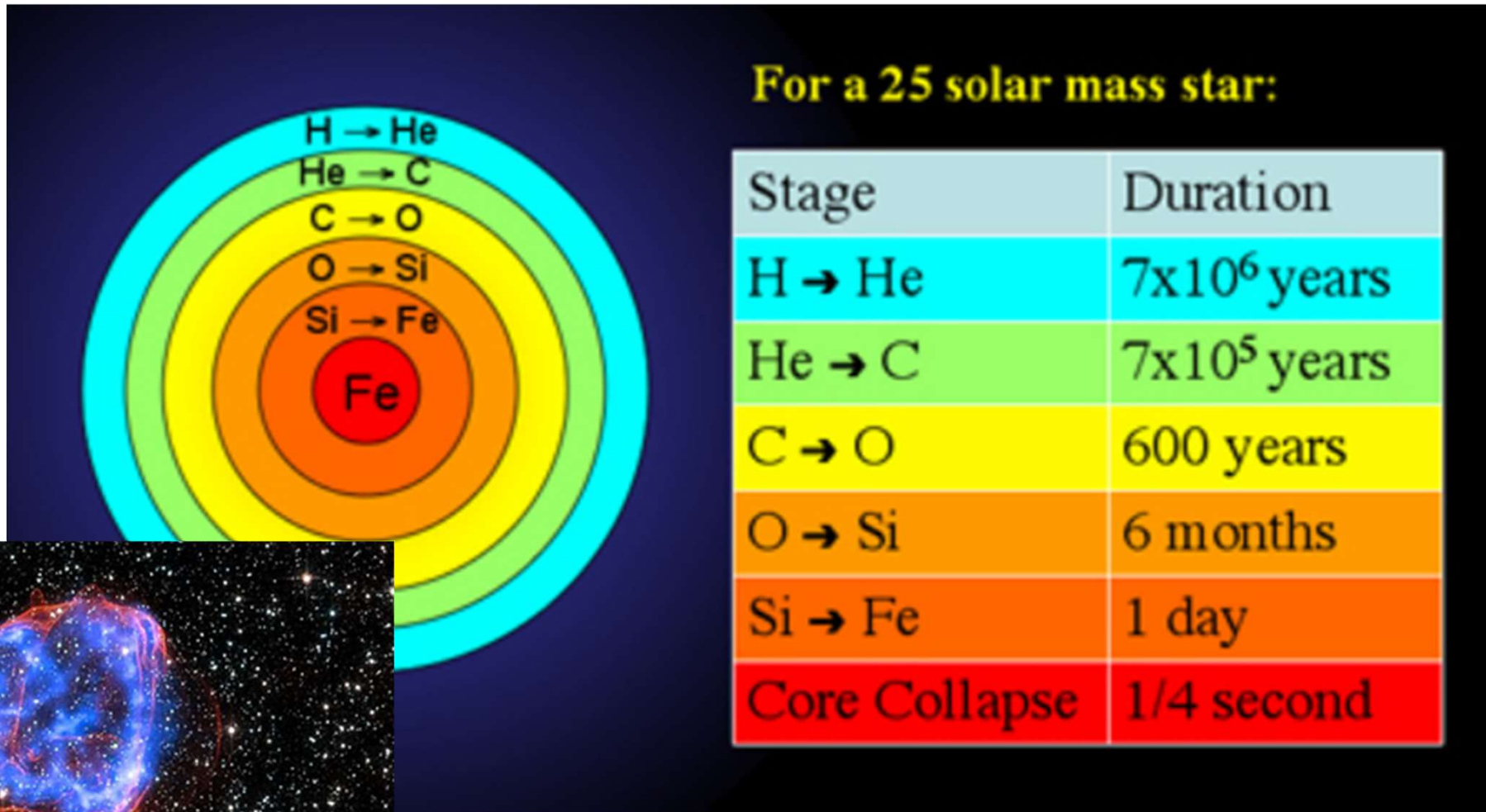
<https://en.wikipedia.org/wiki/Nebula>



<https://en.wikipedia.org/wiki/Supernova>

- The elements H and some He were created in the hot universe during the big bang
- Li, Be and B also formed through the interaction of these with cosmic radiation
- By mass most of the Universe and the Solar system is just H and He.
- Elements till iron were created in stars through fusion.
- Elements beyond Iron can only be created in large stars or supernova explosions via neutron capture or via endothermic processes during the explosion.

The creation of elements in very massive (and short lived) stars



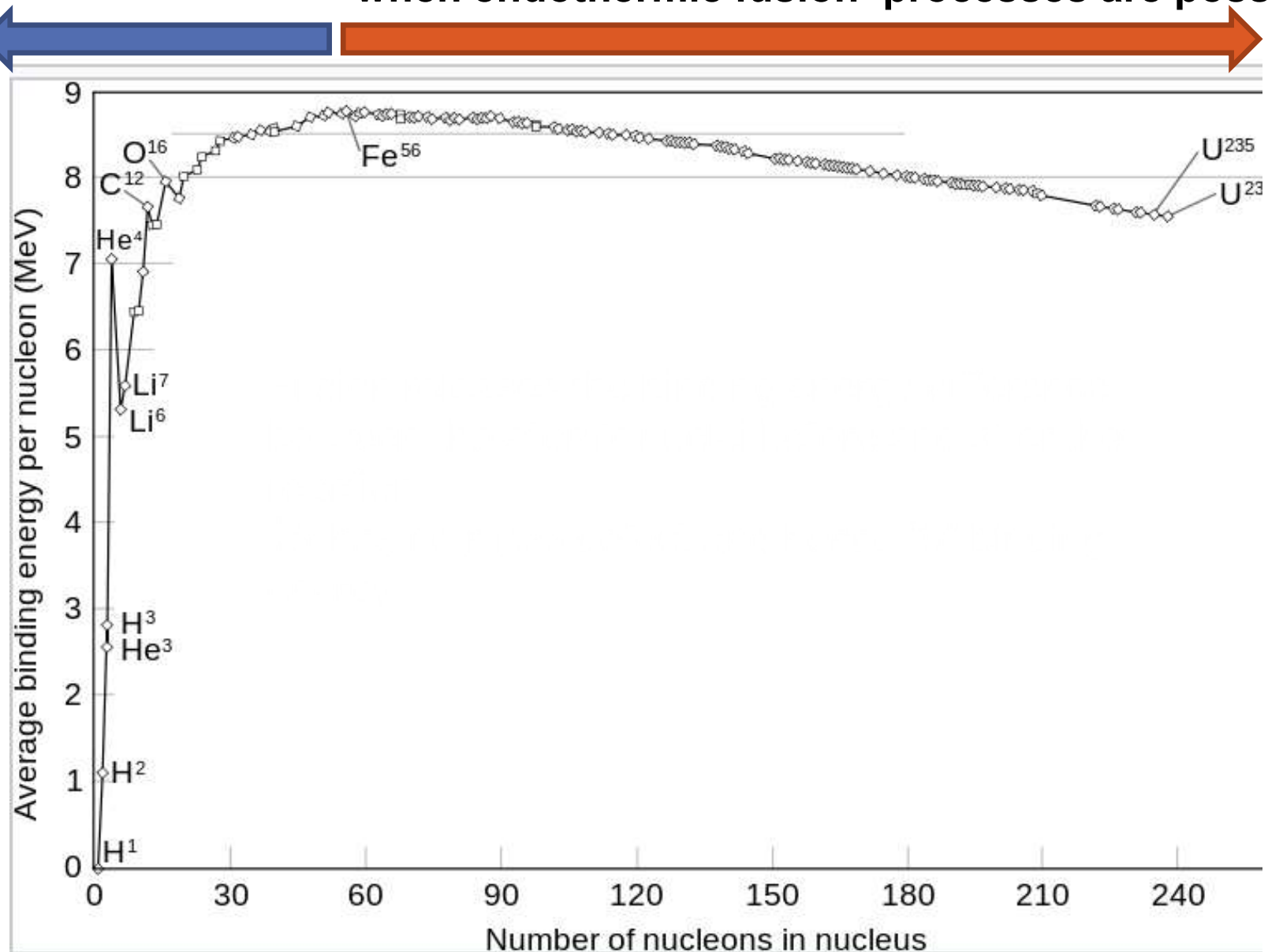
<https://astronomy.swin.edu.au/cosmos/c/core-collapse>

Supernova explosions send clouds of dusts containing heavier elements into the universe. This material is incorporated into new solar systems.

The creation of elements is limited by thermodynamics

Energy can be created through fusion in stars

Energy can be created through fission (We do this for Uranium in a nuclear reactor). Elements are created via neutron capture or during the stars explosion when endothermic fusion processes are possible



The creation of elements

Table 2-1

Solar abundances of the elements (atoms/ 10^6 atoms of Si)

Element	Abundance	Element	Abundance	Element	Abundance
1 H	2.72×10^{10}	29 Cu	514	58 Ce	1.16
2 He	2.18×10^9	30 Zn	1.26×10^3	59 Pr	0.174
3 Li	59.7	31 Ga	37.8	60 Nd	0.836
4 Be	0.78	32 Ge	118	62 Sm	0.261
5 B	24	33 As	6.79	63 Eu	0.0972
6 C	12.1×10^7	34 Se	62.1	64 Gd	0.331
7 N	2.48×10^6	35 Br	11.8	65 Tb	0.0589
8 O	2.01×10^7	36 Kr	45.3	66 Dy	0.398
9 F	843	37 Rb	7.09	67 Ho	0.0875
10 Ne	3.78×10^6	38 Sr	23.8	68 Er	0.253
11 Na	5.7×10^4	39 Y	4.64	69 Tm	0.0386
12 Mg	1.075×10^6	40 Zr	10.7	70 Yb	0.243
13 Al	8.49×10^4	41 Nb	0.71	71 Lu	0.0369
14 Si	1.00×10^6	42 Mo	2.52	72 Hf	0.176
15 P	1.04×10^4	44 Ru	1.86	73 Ta	0.0226
16 S	5.15×10^5	45 Rh	0.344	74 W	0.137
17 Cl	5.24×10^3	46 Pd	1.39	75 Re	0.0507
18 Ar	1.04×10^5	47 Ag	0.529	76 Os	0.717
19 K	3.77×10^3	48 Cd	1.69	77 Ir	0.660
20 Ca	6.11×10^4	49 In	0.184	78 Pt	1.37
21 Sc	33.8	50 Sn	3.82	79 Au	0.186
22 Ti	2.40×10^3	51 Sb	0.352	80 Hg	0.52
23 V	295	52 Te	4.91	81 Tl	0.184
24 Cr	1.34×10^4	53 I	0.90	82 Pb	3.15
25 Mn	9.51×10^3	54 Xe	4.35	83 Bi	0.144
26 Fe	9.0×10^5	55 Cs	0.372	90 Th	0.0335
27 Co	2.25×10^3	56 Ba	4.36	92 U	0.0090
28 Ni	4.93×10^4	57 La	0.448		

The most abundant elements are those created in the Big Bang

Next are those that can be created in normal sized stars (up to ^{16}O).

After that come those than can only be formed in very large stars (up to Fe).

Least are those that can only be formed through neutron capture.

Those that can form via neutron capture in large stars are more abundant than those needing a supernova

The more neutrons they need to catch to form the less abundant they are in general

The creation of elements

H B	B Big Bang L Large stars \$ Super-novae C Cosmic rays S Small stars M Man-made
--------	---

During the red giant phase of a large star certain elements are formed in a neutron rich environment through slow electron capture (s process).

Other elements can only be formed in a supernova explosion formed via rapid neutron capture in a very neutron rich environment, because an unstable isotope with a lifetime in seconds needs to catch the next neutron before it decays (r process)

The creation of elements in red giants

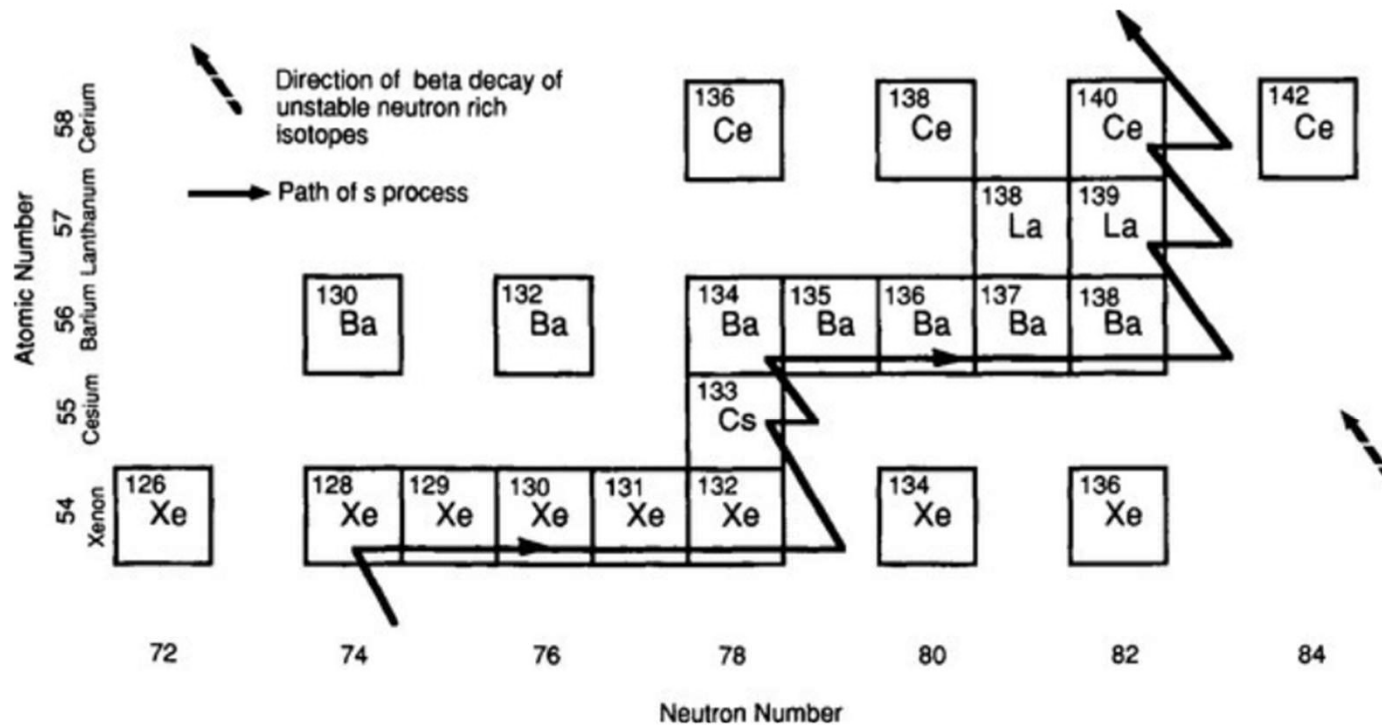


FIG. 2-3 Schematic showing the path of the s process.

The isotopes ^{134}Xe , ^{136}Xe , and ^{142}Ce are beyond the reach of fusion

Star processes do generate neutrons and alpha particles. Starting with a seed isotope successive slow neutron capture reactions can build up an increasingly neutron rich isotope of the same element until an unstable isotope is formed. The unstable isotope decays via beta decay to form the next element. The s process can form elements along the “valley of beta stability” in the chart of nuclides.

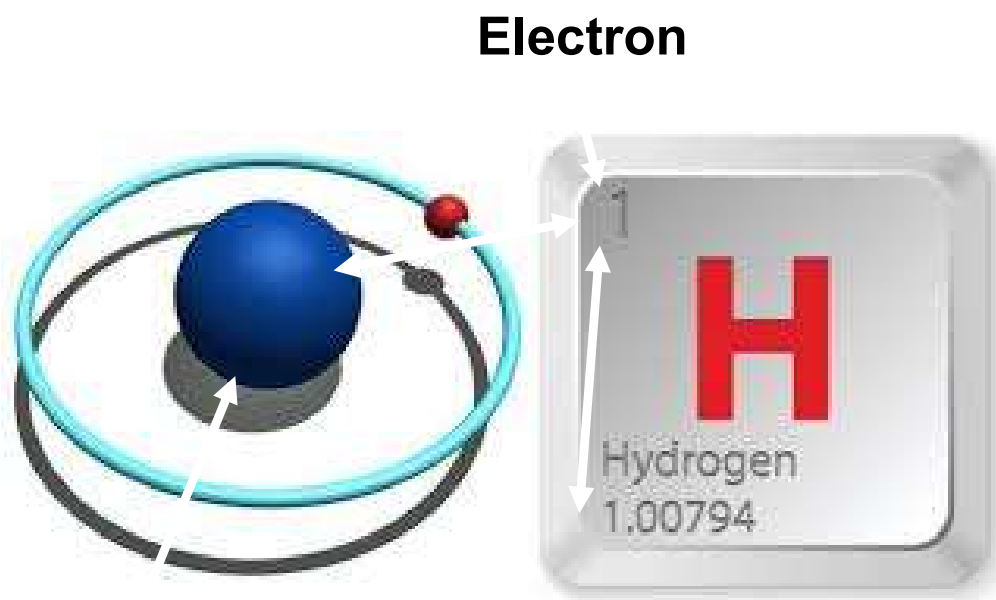
The creation of elements in red giants

Table of Nuclides is arranged differently from the Periodic Table of Elements

Black squares represent stable atoms. Other colours indicate the modes of radioactive decay, e.g. by emission of alpha particles (α), beta particles (β), neutrons (n), etc.

stable	p	α	β β^+	IT	β^-	α	CE	n
--------	---	----------	----------------------	----	-----------	----------	----	---

The structure of atoms



This is how the data can be found in the periodic table. However, when we write down data in the field of mass spectrometry, we write the mass as superscript (A and the number of protons (Z) as subscript (and often omit the later since we are anyway using the elements symbol)

${}^1_1\text{H}$ = Hydrogen with mass 1 amu
 ${}^2_1\text{H}$ = Deuterium mass 2 amu

$$A = Z + N$$

Therefore:

Mass – Atomic number = Number of Neutrons

The Atom is the smallest subunit into which matter of a certain Element can be broken down without changing the element

Facts about the atom

- Size of atoms $1\text{\AA} = 10^{-10}\text{ m}$
- The mass of atoms is expressed in “atomic mass units” amu.
- The amu of ^{12}C is arbitrarily fixed at 12 the mass of all other atoms is determined by comparing it to ^{12}C
- The number of atoms in a mole 6.02252×10^{23} , the Avogadro constant, is arbitrarily fixed such that one mole of ^{12}C weighs 12 g
- Electrons have the mass of $1/1836.1$ of an hydrogen atom and a negative electrical charge of $1.602176 \times 10^{-19}\text{ C}$ (Coulomb) or 1 e (elementary charge)

Facts about the Nucleus

- **Nucleus is 10,000 times smaller than the atom 10^{-14} m or 10^{-4} Å**
- **Density of nuclear matter is 10^{14} g/cm³ such densities may occur in neutron stars or black holes**
- **The nucleus contains a large number of elementary particles which interact with each other and are organized into patterns. The most important ones for our purpose are the protons and neutrons**
- **There are ~1700 different nuclides in the table of nuclides but only 260 of those are stable. These stable nuclides and a few naturally occurring unstable ones make up the Elements in Periodic table**

Table of Nuclides is arranged differently from the Periodic Table of Elements

The Karlsruhe Nuclide Chart

A nuclide chart is a two dimensional representation of the nuclear and radioactive properties of all known atoms. A nuclide is the generic name for atoms characterized by the constituent protons and neutrons. The nuclide chart arranges nuclides according to the number of protons (vertical axis) and neutrons (horizontal axis) in the nucleus. Each nuclide in the chart is represented by a box containing the element symbol and mass number, half-life, decay types and decay energies, etc.

"Magic" numbers

In nuclear physics, a magic number is a number of protons or neutrons (e.g. 2, 8, 20, 28, 50, 82, 126) which give rise to a complete shell in the atomic nucleus. Lead 208 for example, which consists of 82 protons and 126 neutrons, is called "doubly magic" since both the proton and neutron numbers are "magic".

Number of protons (Z)
Number of neutrons (N)

Lead Z=82

Examples of the nuclide box structure

Tl 232 100 t_{1/2} 1.40 × 10¹⁰ a α 4.012, 3.947 t_{1/2} 1.04 × 10⁴ a β 1.37, α 36.6	Ac 226 105 t_{1/2} 28.37 h β 0.8, 1.1 t_{1/2} 2.46 α 5.96, 5.94, 5.94
Ra 226 88 t_{1/2} 14.9 a α 4.8, 4.6 β 0.3, 0.4	Bi 207 83 t_{1/2} 31.55 a β 0.025, 0.064 t_{1/2} 1.702
Ca 135 20 t_{1/2} 1.63 × 10¹⁵ a β 0.3 t_{1/2} 1.787	Rn 219 86 t_{1/2} 3.96 s α 6.819, 6.903 t_{1/2} 6.425 β 0.3, 0.3

Black squares represent stable atoms. Other colours indicate the modes of radioactive decay, e.g. by emission of alpha particles (α), beta particles (β), neutrons (n), etc.

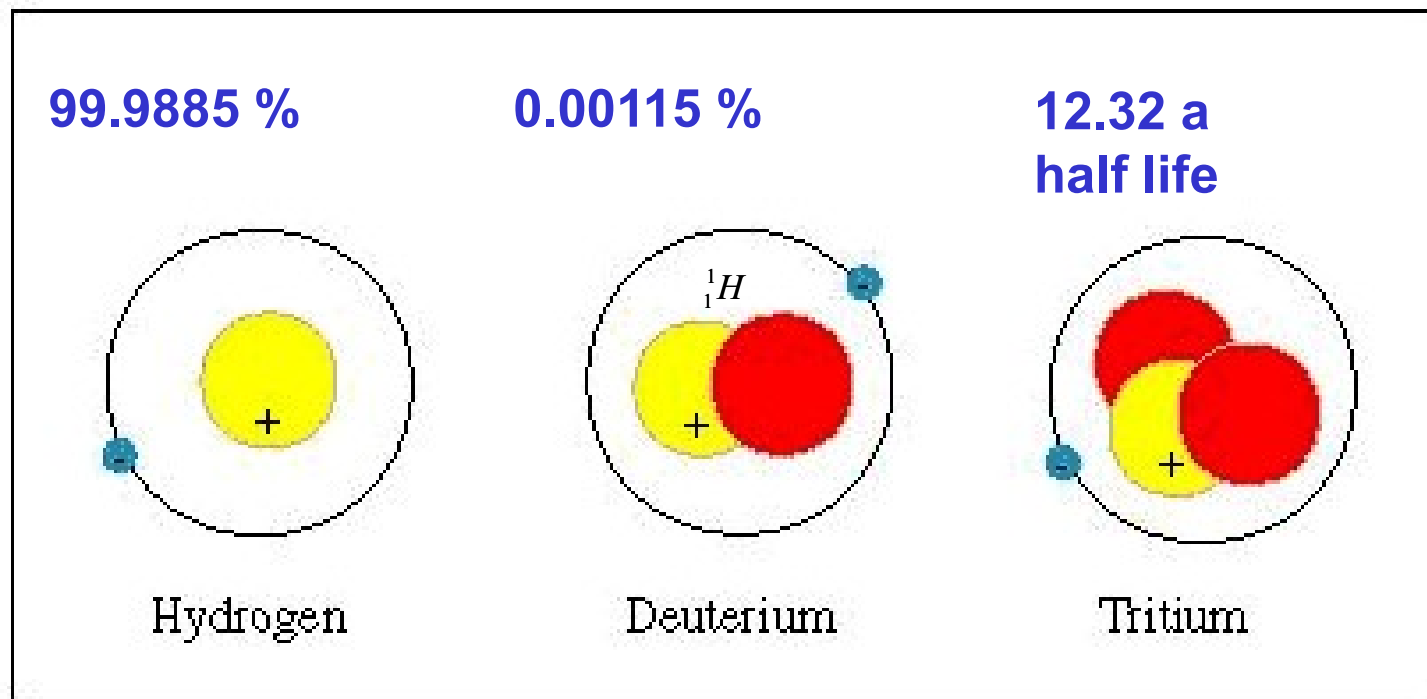
stable	p	α	β ⁺	IT	β ⁻	sf	CE	n
--------	---	---	----------------	----	----------------	----	----	---

Facts about the Protons and Neutrons

- Protons have a positive charge that is equal in magnitude but opposite in polarity to the charge of an electron and together with the electron a mass of 1.007825 amu
- Neutrons have a slightly larger mass (1.008665 amu) and carry no electrical charge. Extra-nuclear neutrons are unstable and decay spontaneously to form protons and electrons with a “half-life” of 10.6 minutes
- Hence neutron capture requires an active neutron source

Isotope

- The name refers to atoms of elements that have **different number of neutrons** in them. **Isotopes** is Greek and means “same place” it refers to atoms that **occupy the same place** in the periodic table of elements but have **different weight**.
- The abundance of stable isotopes is expressed in % in the table of isotopes



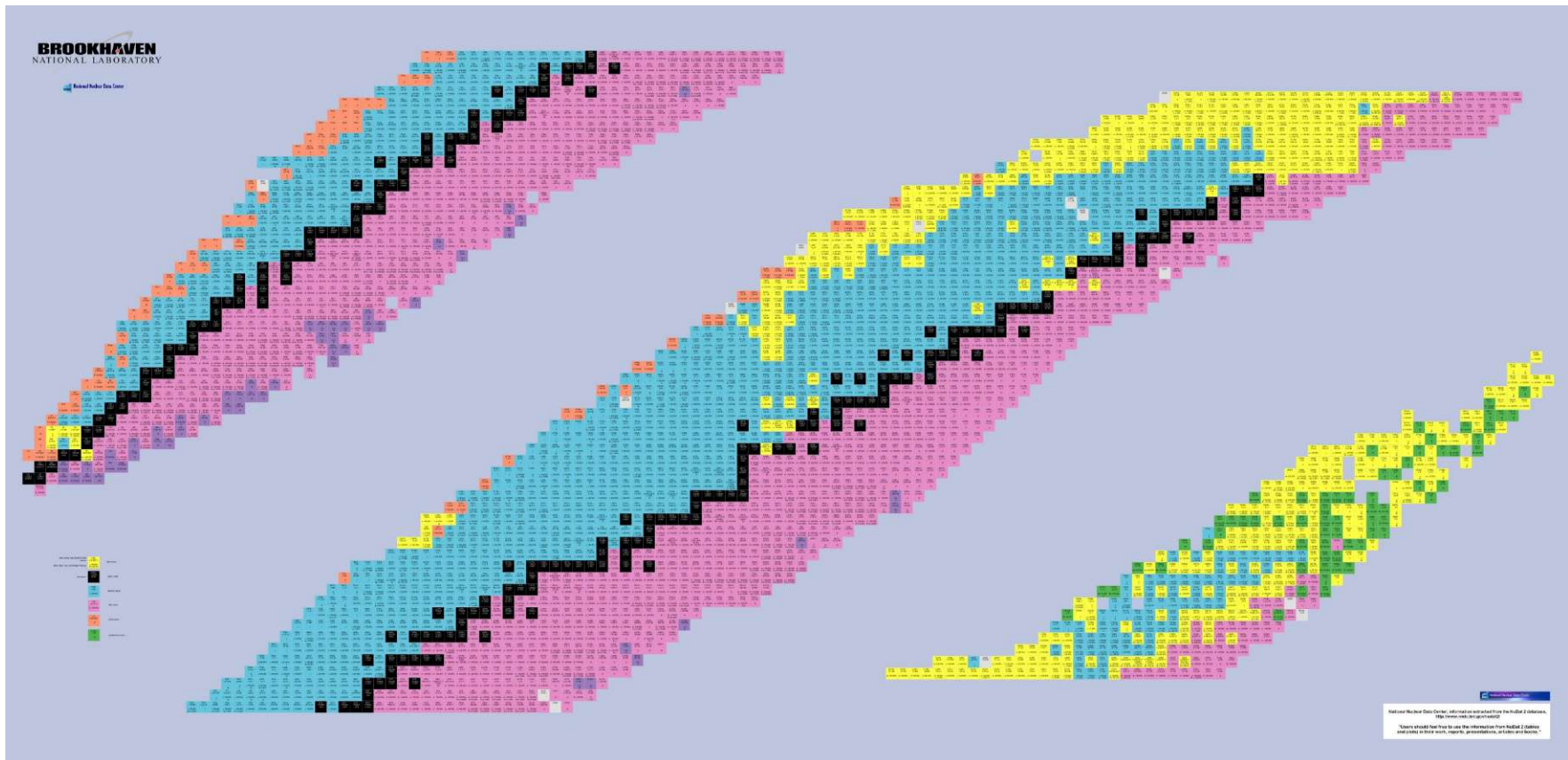
99.9885% ^1_1H means in a sample of 1,000,000 hydrogen atoms 999885 are ^1_1H and 115 are ^2_1H . With this information one can calculate the mass of Hydrogen as $0.999885 \times 1.00782504 + 0.000115 \times 2.014101 = 1.007941$

Elements and Isotope

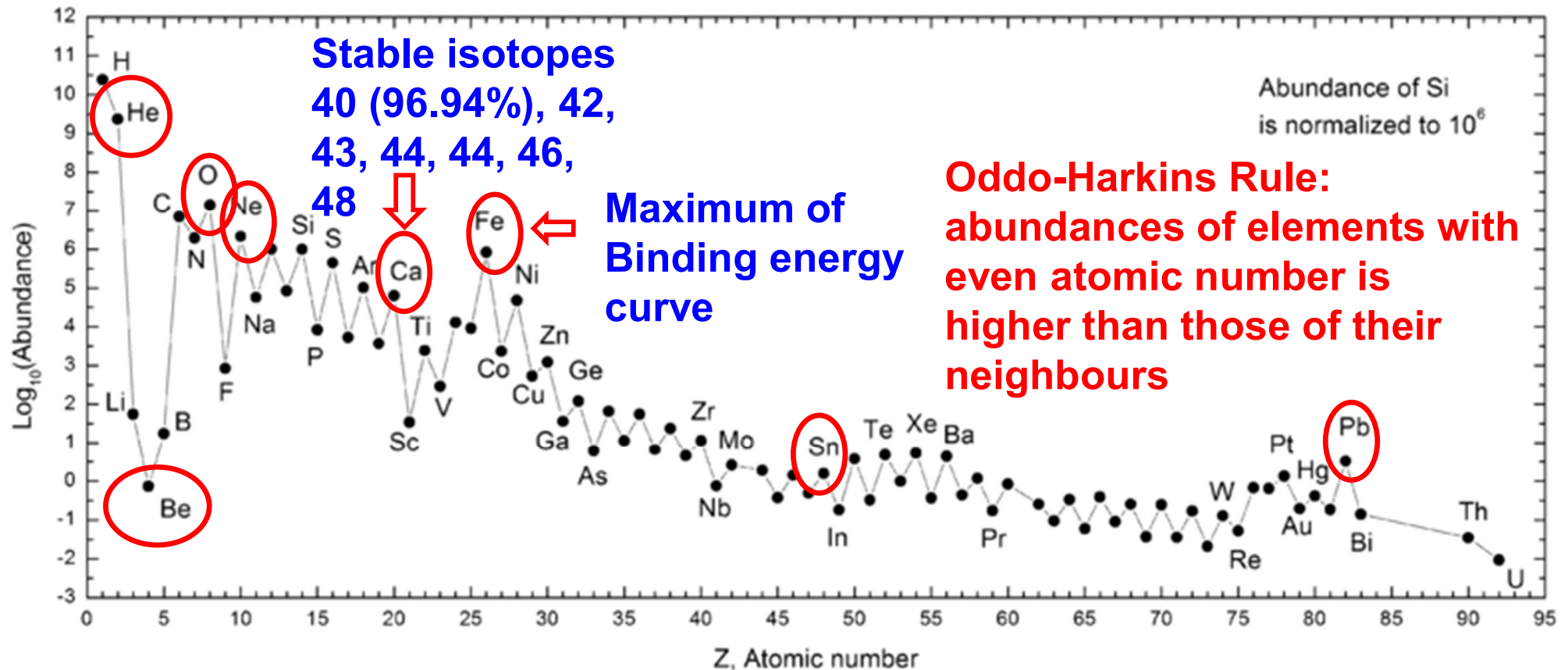
- **Elements:** When two nuclei have different number of protons they belong to different elements. E.g. ^{14}C is an isotope of carbon while ^{14}N is an isotope of nitrogen. One has 6 protons and the other one has 7 protons.
- **Isotopes:** When two nuclei have the same number of protons but different number of neutrons then they are called isotopes. ^{16}O , ^{17}O and ^{18}O are all stable isotopes of oxygen.
- **Radioactive decay:** The vast majority of nuclei formed via fusion or neutron capture are unstable. The decay by releasing alpha, beta or gamma radiation till the nucleus reaches a stable configuration. Radioactive elements and their stable daughters (the decay products that remain behind in rocks and meteorites) tell us a lot about the early solar system and the history of Earth.

Elements and Isotope

- A table of nucleotides shows all the nuclei that have ever been observed in nature or created in the lab via processes like neutron capture and it provides information on how things decay. **Only the black boxes indicate stable isotopes! Stability is the exception not the rule.**



Abundance of elements in the universe

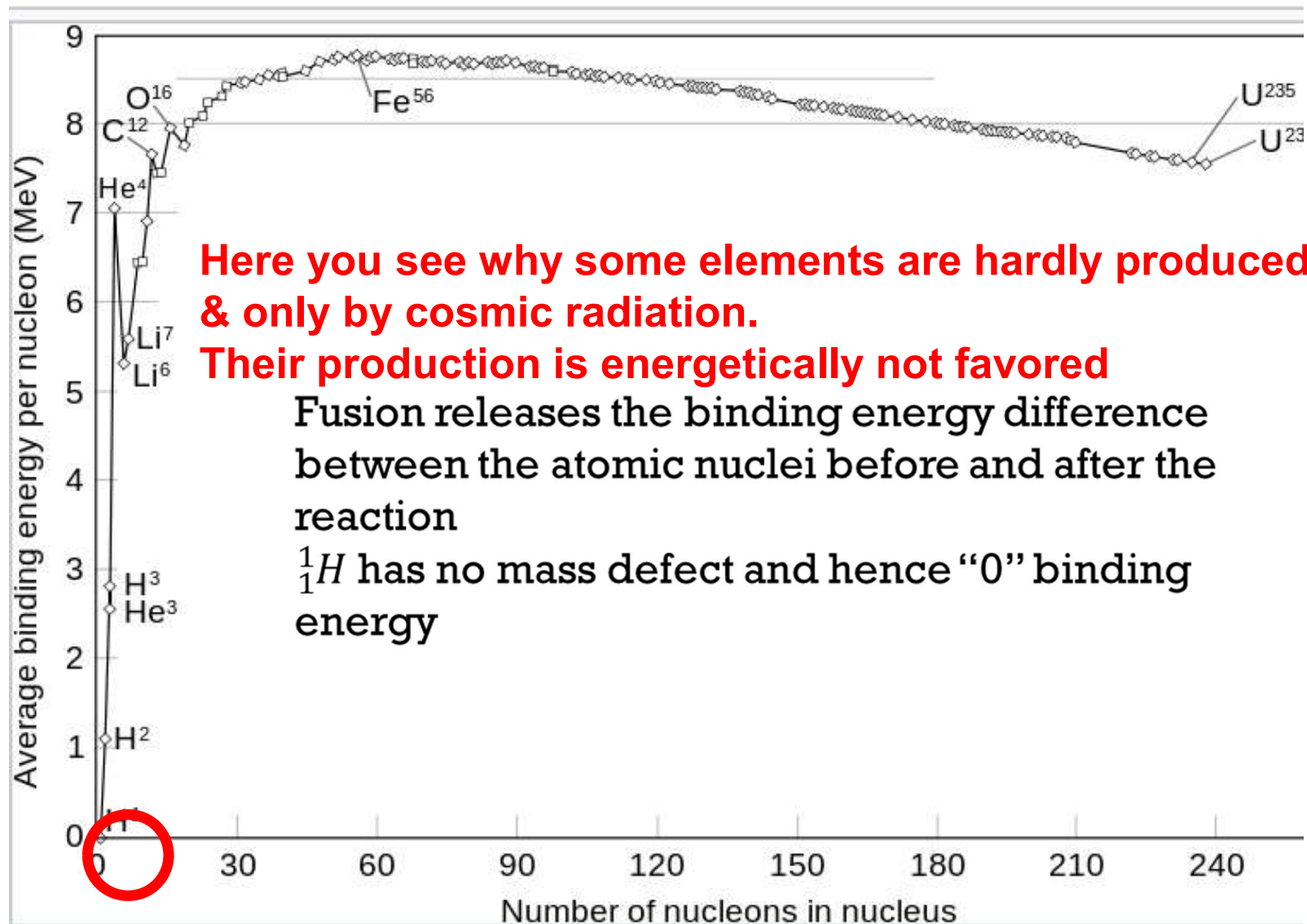


The observed data shows that there is a relationship between nuclear stability and the number of nucleons. Stability is the exception (~260 out of ~1700 known nuclides) not the rule. Small nuclides have N/Z ratios of 1 the largest ones N/Z ratios of 3. Only ^1H and ^3He have more protons than neutrons. There seem to be “Magic numbers” that are more stable than others: 2, 8, 10, 20, 28, 50, 82, and 126. Unstable nuclei decay after their formation till they reach a stable configuration

Distribution of stable nuclides depending on evenness and oddness of A, Z, N

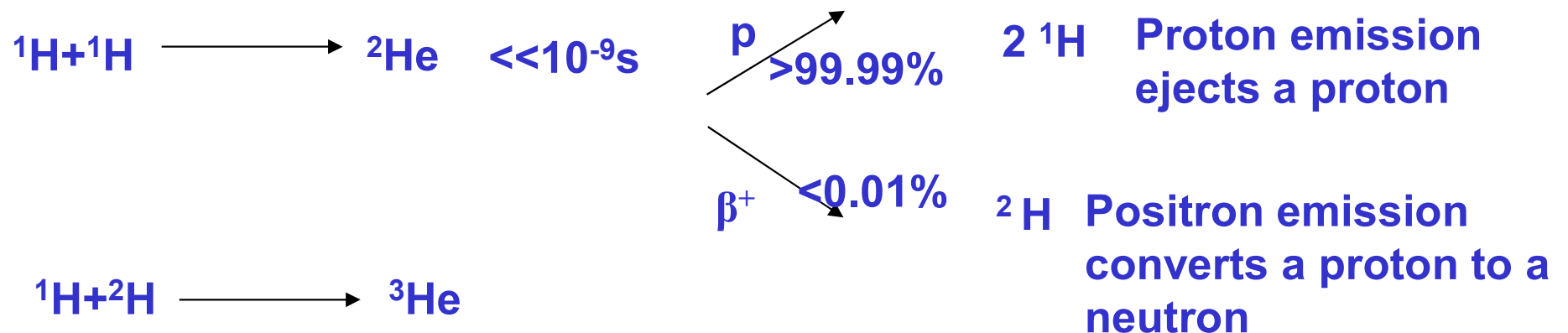
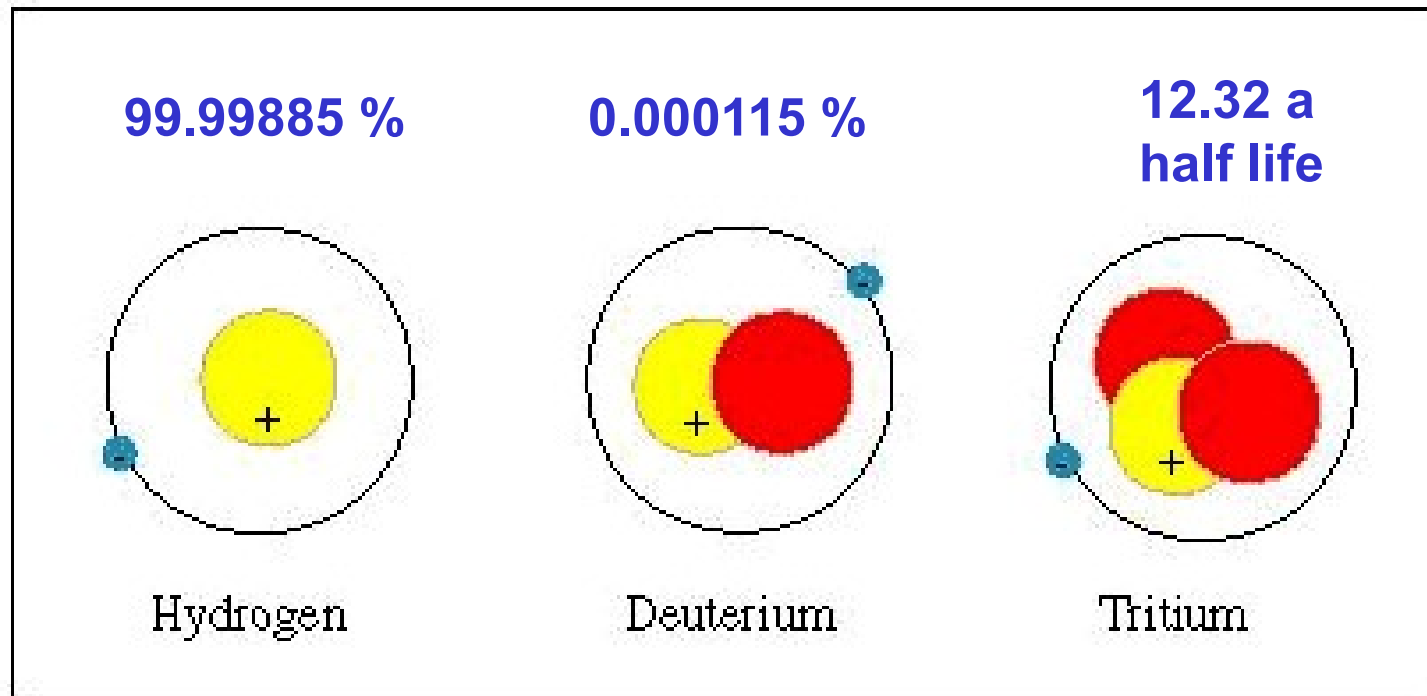
A	Z	N	Number of Stable Nuclides
Even	Even	Even	157
Odd	Even	Odd	53 (Premordial)
Odd	Odd	Even	49
Even	Odd	Odd	4
Odd	Odd		1 (H)
Odd	Odd	Odd	0
			265

How stars can generate energy from Nuclear Fusion



Creating new isotope through fusion

- How to make new elements starting from Hydrogen



Energy released by 2 ^1H to ^2H

- Mass of ^1_1H = 1.007825 amu, Mass of ^2_1H = 2.014101 amu
- Mass of 1 proton + electron + 1 neutron: 1.007825 amu + 1.008665 amu = 2.01649 amu
- $\Delta M = \text{should} - \text{is} = 2.01649 \text{ amu} - 2.014102 \text{ amu} = 0.002388 \text{ amu}$
- Mass of one amu in SI unit follows from the definition of the atomic mass unit: $1 \text{ amu} = 1/12 \times 12/\text{N} = 1/\text{N} \text{ gram}$ where N is the Avogadro's Number.
- Inserting into $E = \Delta M c^2$ yields

$$E [\text{eV}] = (0.002389 \times 10^{-3}) / (6.022141 \times 10^{23}) [\text{kg}] \times (2.997925 \times 10^8)^2 [\text{m}^2 \text{s}^{-2}] / (1.60217653 \times 10^{-19}) = 0.002937 / 6.02252 [\text{kg}] \times (2.997925)^2 [\text{m}^2 \text{s}^{-2}] / 1.60217653 \times 10^9 = 2.22 \times 10^6 [\text{eV}] \text{ or } 1.11 [\text{MeV}] \text{ per nucleon}$$

Our Sun

Our sun is a main sequence star burning Hydrogen to He.
It will stay like that another ~4 Ga eventually it will end as a red giant that turns into a neutron star. Heavier elements like C, N and O are produced in the red giant phase

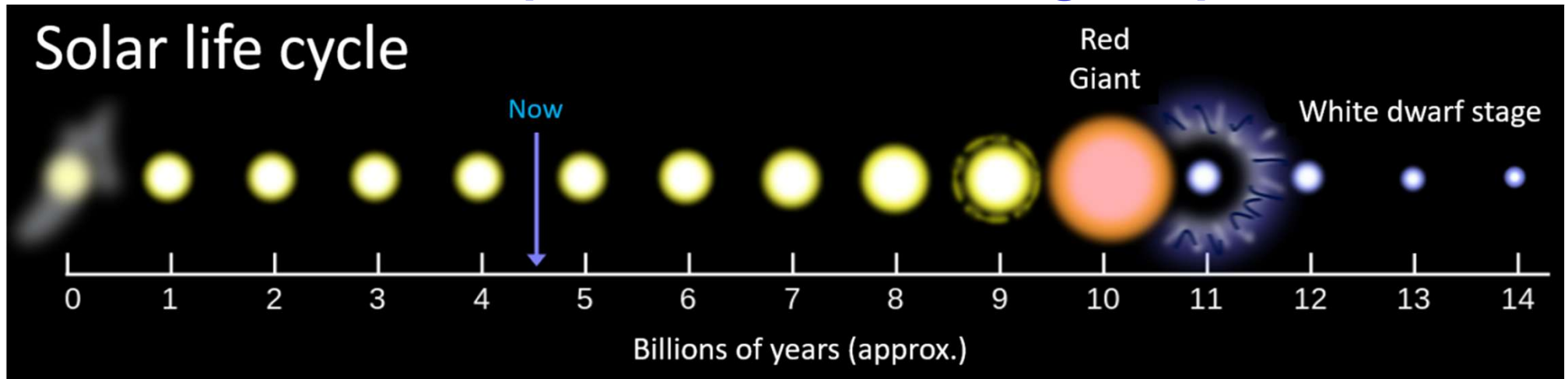


Figure 3.1.1 The Life Cycle of a Main Sequence Star Like Our Sun. Note that the depiction of the diameter of the Sun is not to scale. During the Red Giant phase the Sun is expected to be about 200 times larger than it is now, almost big enough to engulf the Earth, and certainly too close for anything to live here.

<https://environmental-geology-dev.pressbooks.tru.ca/chapter/changes-in-solar-output-and-in-the-earths-atmosphere/>

Our Sun

When a star burns H to He the average mass in the core increases. To maintain hydrostatic equilibrium the star needs to become brighter (burn faster). **Solar luminosity increased by ~ 30% from the Hadean to now. Had this not been compensated by reduction of CO₂ levels in the atmosphere, Earth would be much hotter.**

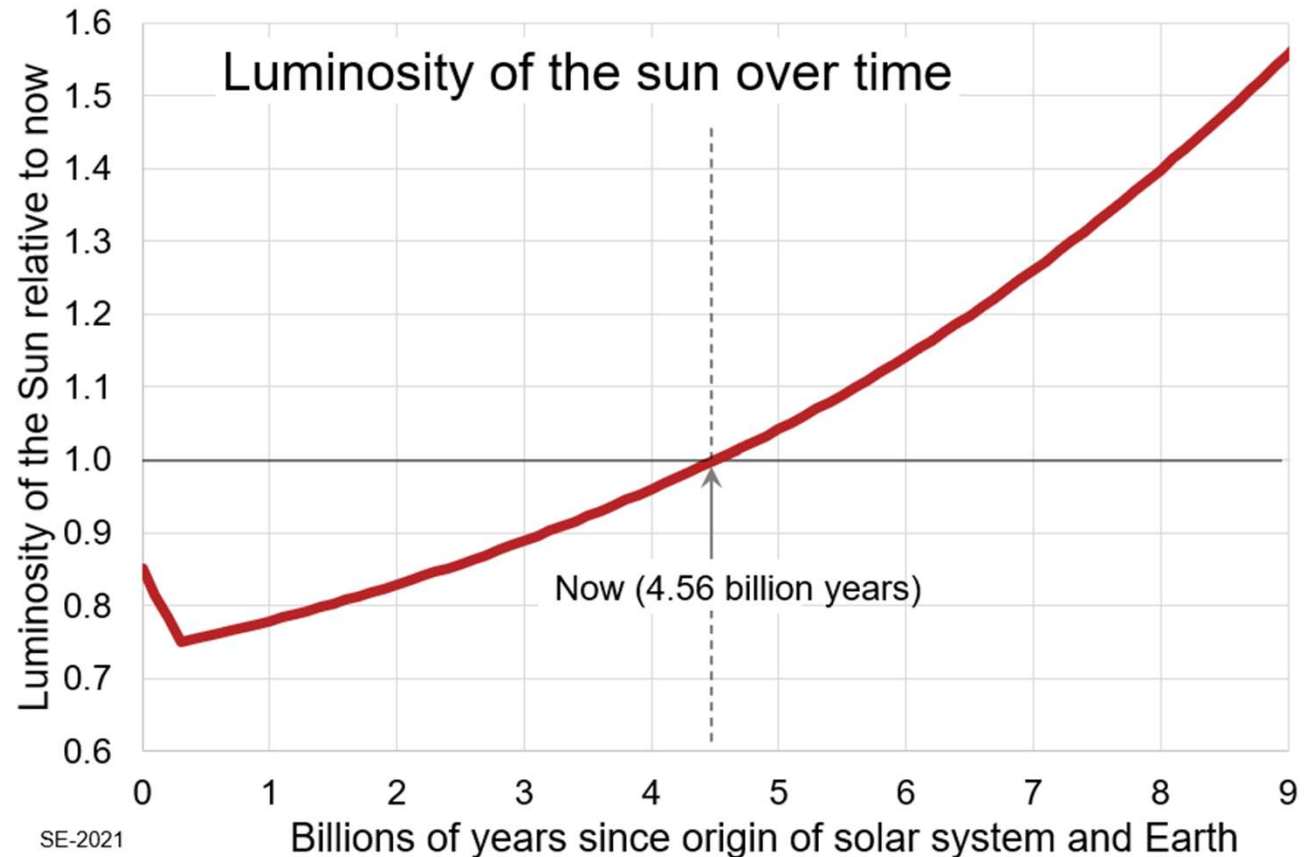


Figure 3.1.2 The Change in Luminosity of Our Sun, and therefore, in the Amount of Solar Energy Received on Earth

<https://environmental-geology-dev.pressbooks.tru.ca/chapter/changes-in-solar-output-and-in-the-earths-atmosphere/>

PRACTICE QUESTIONS

You can submit the answer to these questions by Monday 11th August if you want to get them corrected before the Midsem exam

MCQ style

- 1) **What two things – that are fundamental to our understanding of the world and Universe – began with the Big Bang?**
 - a) Stars and planets
 - b) Gravity and friction
 - c) Space and time
 - d) Atoms and energy
- 2) **Which of the following supports the theory that the Universe is expanding?**
 - a) Radioactive decay occurring in the Sun
 - b) Nuclear fusion occurring in the Sun
 - c) Redshift of light from distant galaxies
 - d) Blueshift of light from distant galaxies
- 3) **Which of the four fundamental forces pulled together hot, dense areas of matter in the early Universe to jumpstart the process of star formation?**
 - a) Weak nuclear force
 - b) Strong nuclear force
 - c) Electromagnetism
 - d) Gravity

MCQ style

- 1) The early Universe was quite uniform. What then were the Goldilocks Conditions for the formation of stars?
 - a) Tiny variations in the density of matter
 - b) Large variations in temperature
 - c) The existence of gravity
 - d) A concentration of heavy metals
- 2) What is the heaviest element that can be created by fusion in a star's core?
 - a) Carbon
 - b) Iron
 - c) Helium
 - d) Silver
- 3) What percent of atomic matter in the Universe is responsible for creating objects more complex than stars?
 - a) 2%
 - b) 42%
 - c) 74%
 - d) 98%

MCQ style

- 1) **Where did most of the elements of the periodic table come from?**
 - a) They came from protons fusing to make elements.
 - b) They came from the Big Bang.
 - c) They came from dying fusion stars.
 - d) They came from dying supernovae.
- 2) **Approximately 98% of the atomic matter in the Universe consists of which of the following?**
 - a) Hydrogen and Helium
 - b) Hydrogen and Oxygen
 - c) Oxygen and Helium
 - d) Silicon and Helium
- 3) **How was Hubble able to calculate a rough estimate for the formation of the Universe?**
 - a) By calculating the age of the oldest atoms
 - b) By calculating when hydrogen was first formed
 - c) By calculating the rate at which the Universe is shrinking
 - d) By calculating the rate at which the Universe is expanding

Numerical questions

1. How many moles of Si are there in 10 g of SiO_2
2. How many protons and neutrons are there in ^{16}O
3. Calculate the binding energy per nucleon for ^{12}C (12 amu)
 $M_{\text{H}} = 1.007825 \text{ amu}$, $M_{\text{N}} = 1.008665 \text{ amu}$, velocity of light
 $2.997925 \times 10^{10} \text{ cm/sec}$, $\text{eV} = 1.60210 \times 10^{-12} \text{ erg}$,
 $N_{\text{A}} = 6.02252 \times 10^{23}$

Long answer style

1. The Big Bang was originally proposed via deductive reasoning from the equations of general relativity. List the key observational evidences supporting the big bang theory.
2. Sketch the reaction how stars burn H into He
3. Explain why elements heavier than Fe can not be created through fusion
4. Explain the difference between elements and isotopes